

HARSHITA KUMARI

Software Developer | Go • Distributed Systems • OCI Registries • P2P Content Distribution

Bengaluru, India | harshitaakri06092002@gmail.com | +91 81023 45588

PROFILE

Software developer focused on Go, cloud-native infrastructure and distributed systems, with hands-on work in content-addressed synchronization, OCI/container caching, peer fallback, concurrency and failure recovery. Built multi-node prototypes that verify transferred content by digest and continue operating through peer or network failures - a strong fit for air-gapped artifact distribution and registry systems.

EXPERIENCE

Software Intern | Matters.ai

Apr 2026 - Jul 2026

Product: DSPM / DDR Data Security Platform | Bengaluru, Karnataka

- Built and debugged backend components and API integrations for a data-security platform, translating product requirements into reliable service behavior.
- Worked on data-flow and synchronization paths with emphasis on failure handling, repeatable debugging, and safe state transitions across service boundaries.
- Contributed in a Git-based engineering workflow across implementation, testing, integration, and iterative fixes.

Research Intern | IIT Madras

Jul 2025 - Dec 2025

Product: Distributed Systems Research Platform | Chennai, Tamil Nadu

- Built distributed-systems prototypes and reproducible multi-node experiments to study behavior under partial failures, retries, and changing node availability.
- Analyzed system behavior, isolated failure modes, and iterated on implementations from observed results rather than assuming ideal network conditions.
- Documented experiment design and results to make debugging and performance comparisons repeatable.

Backend Developer Intern | Celebal Technologies

May 2024 - Jul 2024

Product: Enterprise Data Platform - Backend Services

- Developed backend services and API-driven data flows, with attention to service integration, request handling, and correctness across failure scenarios.
- Debugged and tested backend behavior across multiple scenarios and used collaborative Git workflows to ship incremental changes.

SELECTED PROJECTS

Distributed File Synchronization Tool | Personal Project

Go | Docker | Linux | SHA-256 | REST | Concurrency

- Built a peer-to-peer synchronizer that splits files into content-addressed chunks, transfers only missing chunks, and verifies every chunk by SHA-256 after transfer.
- Added concurrent workers, retries, and peer failover so synchronization can continue when a source node becomes unavailable.

Container Image Cache with Peer Fallback | Personal Project

Go | OCI | Docker | REST | Docker Compose

- Built a multi-node image cache with local-cache → peer/primary → fallback-registry resolution; verified blobs by digest before making them available.
- Prevented incomplete downloads from entering the cache and implemented eviction, concurrency handling, and multi-node testing with Docker Compose.

Fault-Tolerant Distributed Job Queue | Personal Project

Go | REST | Docker | Concurrency

- Implemented concurrent workers, lifecycle tracking, retries, worker-failure detection, and duplicate-processing protection.

TECHNICAL SKILLS

Core: Go, Python, Linux, REST APIs, Git/GitHub, CI/CD

Distributed Systems: P2P systems, concurrency, fault tolerance, retries, peer failover, content-addressed storage, digest verification

Cloud Native / Registry: OCI artifacts & registries, Docker, Docker Compose, container caching, image distribution

EDUCATION

B.Tech, Computer Science & Engineering | Galgotias University, Greater Noida

CGPA: 9.0